Nithin C Babu

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Research Interests

- **Visual Quality Assessment**: Machine learning methods for assessing user/AI-generated content.
- **Deep Learning**: Image or video processing
- **Computer Vision**: Feature extraction, scene understanding

Research Publications

- Nithin C Babu, Aniruddha Mahapatra, Harsh Rangwani, Kuldeep Kulkarni, and Rajiv Soundararajan. Efficient adaptation of ai-generated video quality assessment with quality descriptions. Manuscript, 2026
- Anatap Mitra*, Nithin C Babu*, Karimulla Shaik, and Rajiv Soundararajan. Text-guided Q-Former for data-efficient ai-generated image quality assessment. Manuscript, 2026
- Nithin C Babu, Aniruddha Mahapatra, Harsh Rangwani, Rajiv Soundararajan, and Kuldeep Kulkarni. DynamicEval: Rethinking evaluation for dynamic text-to-video synthesis. *arXiv preprint:2510.07441*, 2025
- Aakanksha Bharti, Nithin C. Babu, M. Naveen Sai Kiran, Aditi Prasad, Neeraj Badal, and Rajiv Soundararajan. Quality assessment of multispectral remote sensing images: A subjective study and no reference algorithms. *IEEE Transactions on Geoscience and Remote Sensing (TGRS)*, 63:1–16, 2025
- Sanjot Sagar Totade*, Nithin C Babu*, Shika Rao, and Rajiv Soundararajan. Internal embeddings of multi-modal llms as generalizable representations for image quality assessment. In *Proceedings of the Fifteenth Indian Conference on Computer Vision Graphics and Image Processing (ICVGIP)*, January 2025
- Vignesh Kannan, Sameer Malik, Nithin C. Babu, and Rajiv Soundararajan. Quality assessment of low-light restored images: A subjective study and an unsupervised model. *IEEE Access*, 11:68216–68230, July 2023
- Nithin C. Babu*, Vignesh Kannan*, and Rajiv Soundararajan. No reference opinion unaware quality assessment of authentically distorted images. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pages 2459–2468, January 2023

* *Equal contribution*

Research Experience and Internships

- **Adobe India, Bangalore** — Bangalore, Karnataka
  Ph. D. Research Intern — *May 2024 - August 2024*
  - **Quality Evaluation Metrics for AI-Generated Videos**: Research Intern for three months. Focused on evaluating the quality of AI-generated videos through subjective experiments and developing quality metrics that correlate well with human perception.
- **Elvicto Technologies Private Limited, Technopark, Trivandrum** — Trivandrum, Kerala
  Project Intern — *July 2018*
  - **Smart Sensor Design based on LoRaWAN Technology**: Project Intern for one month in designing and developing Smart Sensors based on IoT and LoRaWAN Technologies for Smart City developments.

- **National Institute of Electronics and Information Technology, Calicut** Calicut, Kerala
  Intern *June 2017 - July 2017*
  - **Embedded System Design using Raspberry Pi Internship**: Completed a one-week internship in Embedded System Design and IoT on Raspberry Pi.
- **Bharat Sanchar Nigam Limited, Thrissur** Thrissur, Kerala
  Industrial Trainee *June 2016 - July 2016*
  - **Telecom Technology Industrial Training**: Completed a two-week industrial training on 'Telecom Technology'.

Education

- **Indian Institute of Science, Bangalore** Bangalore, Karnataka
  Ph. D. in Electrical Communication Engineering *October 2020 - Present*
  - Research Advisor : Dr. Rajiv Soundararajan
  - Research Topic : Visual Quality Assessment of User-Generated and AI-Generated Content
  - CGPA : 9.0/10
- **Government Engineering College, Thrissur** Thrissur, Kerala
  Bachelor of Technology in Electronics and Communication Engineering *Aug 2015 - July 2019*
  - Project Advisor : Prof. Anish Babu K K
  - CGPA : 8.03/10
- **Kendriya Vidyalaya Thrissur** Thrissur, Kerala
  CBSE Senior Secondary Examination *2015*
  - Steam : Computer Science
  - Percentage : 94.4%
- **Kendriya Vidyalaya Thrissur** Thrissur, Kerala
  CBSE Secondary Examination; CGPA: 10/10 *2013*
  - CGPA : 10/10

Academic Projects

- **Adversarial attacks on Perceptual Quality Metrics trained on Authentically Distorted Images and Defense strategies**: *June 2021*
  - ***Advanced Image Processing Course Project***: Analyzed the effect of adversarial attacks on deep-feature based perceptual quality metrics and employed adversarial training to improve performance the of deep-features.
- **Learning to See in the Dark**: *January 2021*
  - ***Digital Image Processing Course Project***: Analysis of the paper, 'Learning to See in the Dark' by *Chen et al.* (CVPR 2018). Used the implementation to learn the basics of deep learning and its implementation in Python using Tensorflow. Analyzed the trained network and solved a limitation of the existing paper.
- **Automated Locker Management System for Laundry Services**: *2018 - 19*
  - ***B.Tech. Final Year Project***: Created a fully automated, scalable locker system for Black Swan Dry Cleaners company. Here, the customer can drop off his/her garments in the locker and pick them up when finished, being notified by an SMS to his/her mobile phone.

Academic Experience

- **Indian Institute of Science**:
  - **Main teaching assistant**: E9 241 - Digital Image Processing *August 2022 - December 2022*
    - Involved in setting up and evaluating assignments.
    - Conducted tutorials on Python / MATLAB coding.
  - **Main teaching assistant**: E9 241 - Digital Image Processing *August 2023 - December 2023*
    - Conducted an additional Python / MATLAB vectorization and broadcasting tutorial.

Technical Skills

- **Programming Languages**: Python, MATLAB, C++
- **Deep Learning and Data Science**: PyTorch, NumPy, Pandas
- **Operating Systems**: Windows, Linux

Academic Service

- **Reviewer**:
  - The Fortieth Annual Conference on Neural Information Processing Systems (NeurIPS), 2026
  - ACM International Conference on Multimedia (ACM-MM), 2026
  - European Conference on Computer Vision (ECCV), 2026
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025 **(Outstanding Reviewer)**
  - AAAI Conference on Artificial Intelligence (AAAI), 2025
  - International Conference on Signal Processing and Communications (SPCOM), 2024
  - IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024

Honors and Awards

- Recognized as one of the Outstanding Reviewers for CVPR'25
- Recipient of Prime Minister Scholarship Scheme 2015-'19 for all four years of B.Tech.
- 0.1% Merit in Mathematics in CBSE Senior Secondary Examination

Test Scores

- Graduate Aptitude Test in Engineering (GATE) 2020 - Score 933/100 (AIR 12 in ECE)